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Learning When to Trust via Selective Context Preference Optimization

MIST Benchmark and SCOPE Training Teach Language Models to Resist Misleading Context Without Ignoring Helpful Signals

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A new paper introduces MIST, a benchmark that renders each reasoning item under four matched context conditions, and SCOPE, a DPO-based training method that teaches models to distinguish misleading from helpful context. SCOPE significantly reduces the fraction of correct answers flipped wrong by misleading signals while preserving the model’s benefit from genuinely useful context.

AT A GLANCE

Problem: LLMs are sycophantic toward misleading context yet training to resist context can make them ignore helpful signals.

Why it matters: RAG, web search, and chat history all deliver context that may be wrong; models that cannot discriminate are unsafe to deploy.

Method: MIST benchmark (4 matched conditions per item) + SC2W metric + SCOPE (DPO over balanced preference pairs across all four conditions).

Result: SCOPE significantly reduces SC2W while preserving correct-context utilization; balanced training is crucial.

Evidence: Ablation confirms that optimizing only on misleading pairs degrades helpful-context performance.

The Big Picture

Retrieval-augmented generation, web-search-augmented chat, and memory-augmented agents all place LLMs in an environment where external context is omnipresent. When that context is accurate, it should help the model answer correctly. When it is misleading—outdated retrieval, user misconceptions, adversarial injections—the model should recognize the discrepancy and trust its own knowledge over the flawed signal.

Current LLMs fail in both directions: they are sycophantic toward user-provided context (a misleading signal flips a correct answer wrong), yet attempts to harden models

against misleading context often produce models that ignore helpful context universally. This paper frames the problem as a **discrimination task**: the model must learn to distinguish the quality of context, not merely resist all of it.

Why Existing Approaches Fall Short

Prior robustness work has largely attacked misleading context as a single phenomenon, training on adversarial examples where context is always wrong. This produces models with one failure mode traded for another: high resistance to misleading context, but equally high resistance to correct context, because the model has learned “context → ignore,” not “misleading context → ignore.”

Evaluating this nuance requires a benchmark that explicitly presents both misleading and correct context for the same item. Without matched conditions, it is impossible to separate context-resistance from context-utilization, and prior evaluations have conflated the two.

Core Mechanism

MIST Benchmark. Each item in MIST appears under four matched conditions:

1. **Clean:** question alone, no external context
2. **Misleading:** context that nudges toward the wrong answer
3. **Correct context:** context that supports the right answer
4. **Irrelevant:** context unrelated to the question

The **SC2W metric** (Susceptibility to Context-induced Wrong answers) counts how often condition 2 flips a clean-correct answer to wrong:

$$\text{SC2W} = \frac{\#\{\text{clean-correct} \wedge \text{misleading-wrong}\}}{\#\{\text{clean-correct}\}}$$

SCOPE Training. SCOPE (Selective Context Preference Optimization) applies DPO over preference pairs mined from MIST failures. The preferred completion is the model’s clean-correct answer; the dispreferred completion is the misleading-flipped wrong answer. Crucially, pairs are sampled **equally across all four conditions**, not only misleading ones.

Technical Formulation

Let x be a question and c a context drawn from condition set $\mathcal{C} = \{\emptyset, \text{mis}, \text{cor}, \text{irr}\}$. The DPO objective over preference pairs (y_w, y_l) is:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,c,y_w,y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x, c)}{\pi_{\text{ref}}(y_w | x, c)} - \beta \log \frac{\pi_{\theta}(y_l | x, c)}{\pi_{\text{ref}}(y_l | x, c)} \right) \right]$$

where $\mathcal{D}$ is the balanced dataset spanning all four conditions with equal weight. The SC2W constraint requires that:

$$\mathbb{P}[y^* \mid x, c_{\text{mis}}] > \mathbb{P}[y^* \mid x, c_{\text{mis}}] \Big|_{\pi_{\text{ref}}}$$

while preserving:

$$\mathbb{P}[y^* \mid x, c_{\text{cor}}] \approx \mathbb{P}[y^* \mid x, c_{\text{cor}}] \Big|_{\pi_{\text{ref}}}$$

Learning or Inference Procedure

Pair mining: Identify items where the base model is correct in the clean condition but incorrect in the misleading condition. Each such item yields one DPO preference pair.

Balanced sampling: Sample equal numbers of preference pairs from each of the four conditions. This ensures the model cannot learn a context-agnostic “ignore all signals” policy.

DPO fine-tuning: Standard DPO applied to the preference dataset; no modifications to the DPO algorithm are required.

Inference: No change. The trained model processes queries with external context in the standard RAG pipeline.

What the Guarantee Says

No formal guarantee. The balanced training design provides an intuitive argument: because pairs from the correct-context condition reward trusting helpful context while pairs from the misleading condition reward resisting bad context, the model must learn the discrimination rather than a blanket policy. The empirical ablation confirms this: training on misleading-only pairs degrades correct-context utilization, while balanced training does not.

Experimental Findings

- **SC2W reduction:** SCOPE significantly reduces the fraction of correct answers flipped wrong by misleading context.
- **Correct-context utilization:** Preserved after SCOPE training; the model continues to benefit from genuine context.
- **Irrelevant-context:** Model correctly learns to ignore irrelevant signals.
- **Clean accuracy:** Maintained; SCOPE does not degrade the model’s baseline reasoning.

Ablations and Interpretation

Misleading-only DPO vs. SCOPE. Training DPO only on misleading-context pairs achieves strong SC2W

reduction but degrades correct-context utilization by a significant margin, confirming the “context-ignore” failure mode. SCOPE’s balanced sampling resolves this.

Effect of condition balance. Increasing the proportion of misleading pairs toward 100% monotonically trades SC2W improvement for correct-context degradation. The balanced 50/50/50/50 split across conditions achieves the best combined performance.

MIST item difficulty. High-SC2W items tend to involve superficially plausible misleading contexts (wrong dates, plausible-but-incorrect numbers). These are the cases where the model most needs the discrimination capability SCOPE provides.

Reference

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